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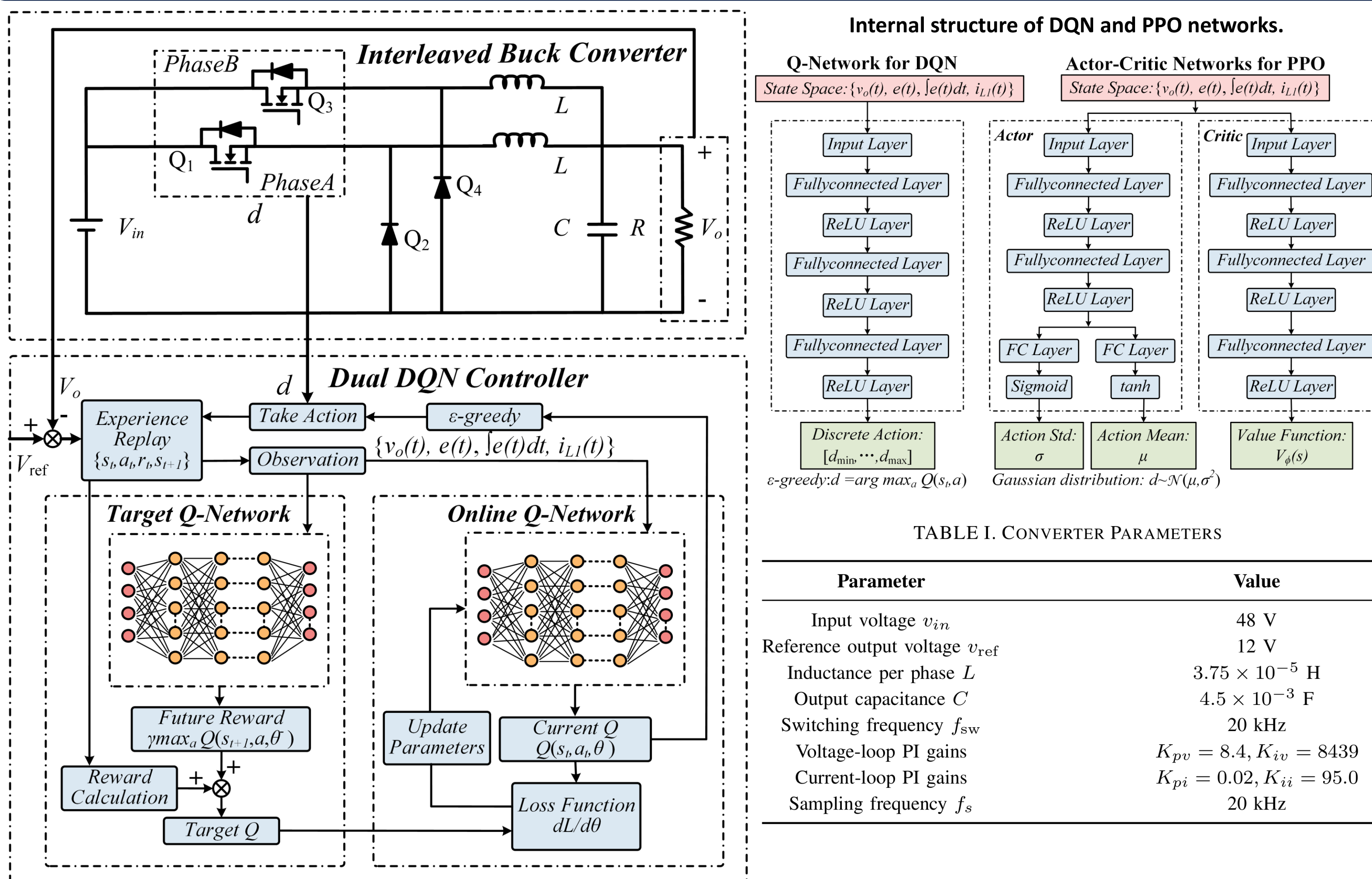
Interleaved buck converter; Transfer learning;

0. Summary

Challenges: Classical model-based control approaches may not readily accommodate parameter drift. When system parameters change, the controller parameters typically need to be redesigned for the new plant. Practical deployment remains challenging due to the computational complexity of deep networks, which may hinder implementation on resource-constrained embedded controllers.

Contribution: This paper presented a model-free DRL-based control framework for multiphase interleaved buck converters. DQN and PPO controllers were developed in simulation and both outperformed PI under load variations. Transfer learning enabled rapid adaptation to inductance variations. A lightweight DQN agent was deployed on an MCU and validated on an HIL platform, demonstrating embedded feasibility.

1. Proposed DRL Control Method



- A model-free DRL controller is proposed for a two-phase interleaved buck converter. The control task is formulated as a Markov Decision Process.
- The agent observes $\{v_o, e, \int edt, i_{L1}\}$ and generates the duty-cycle command directly. DQN is designed for discrete duty-cycle actions, while PPO is designed for continuous duty-cycle output.
- The DQN agent employs a Q-network to approximate the action-value function. The PPO agent adopts an actor–critic architecture, comprising a value (critic) network and a policy (actor) network.

2. Agent Training and Transfer Learning

TABLE II. DQN HYPERPARAMETERS

Parameter	Value
Learning rate α	0.001
Discount factor γ	0.9
Replay buffer size B	2×10^5
Mini-batch size b	256
ε -greedy rate ε	0.1
Action range $[d_{\min}, d_{\max}]$	$[0.22, 0.44]$
Number of actions n	12
Sampling frequency f_s	20 kHz

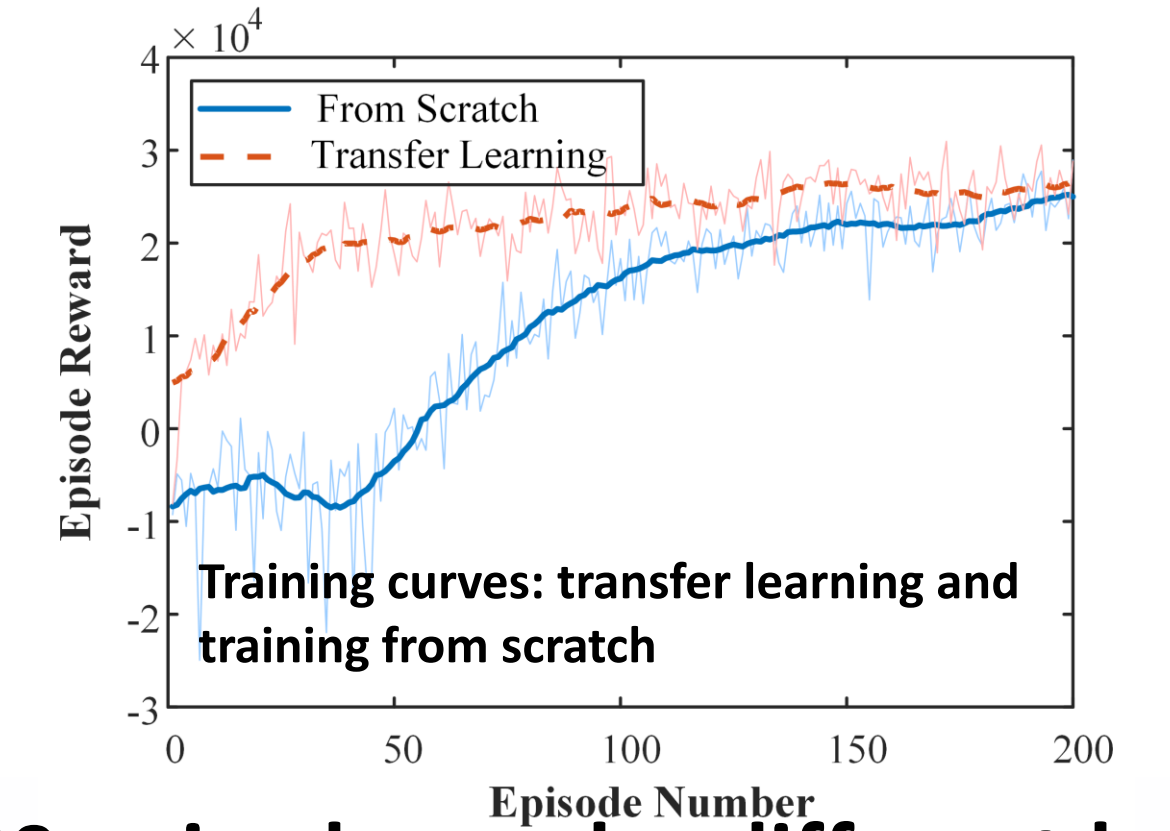
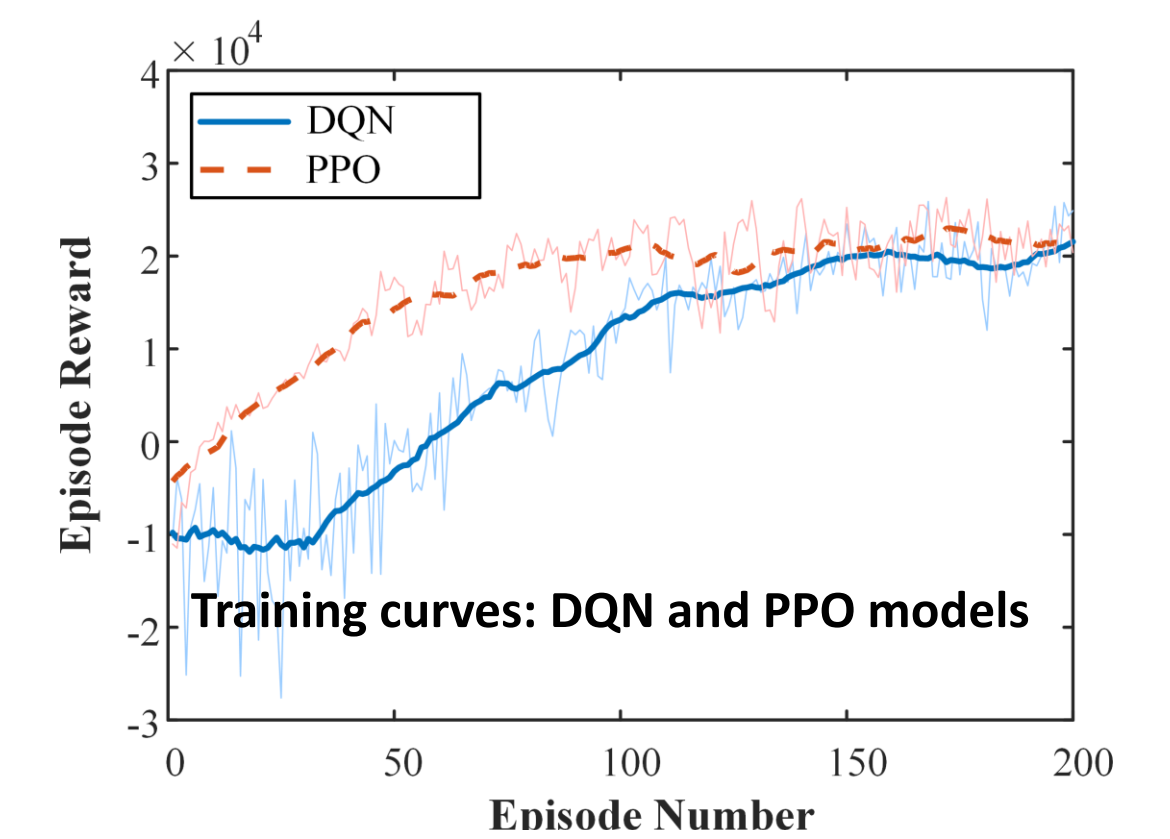
TABLE III. PPO HYPERPARAMETERS

Parameter	Value
Learning rates	$\alpha_{critic} = 10^{-3}$, $\alpha_{actor} = 10^{-4}$
Discount factor γ	0.9
Experience horizon H	256
Mini-batch size b	256
Epochs per update N	5
Entropy weight η	0.02
Clipping factor δ	0.2
GAE factor λ	0.98
Action mean range μ	[0.21, 0.45]
Action std range σ	[0.01, 0.1]
Sampling frequency f_s	20 kHz

Algorithm 1 PPO-Clip Training Procedure

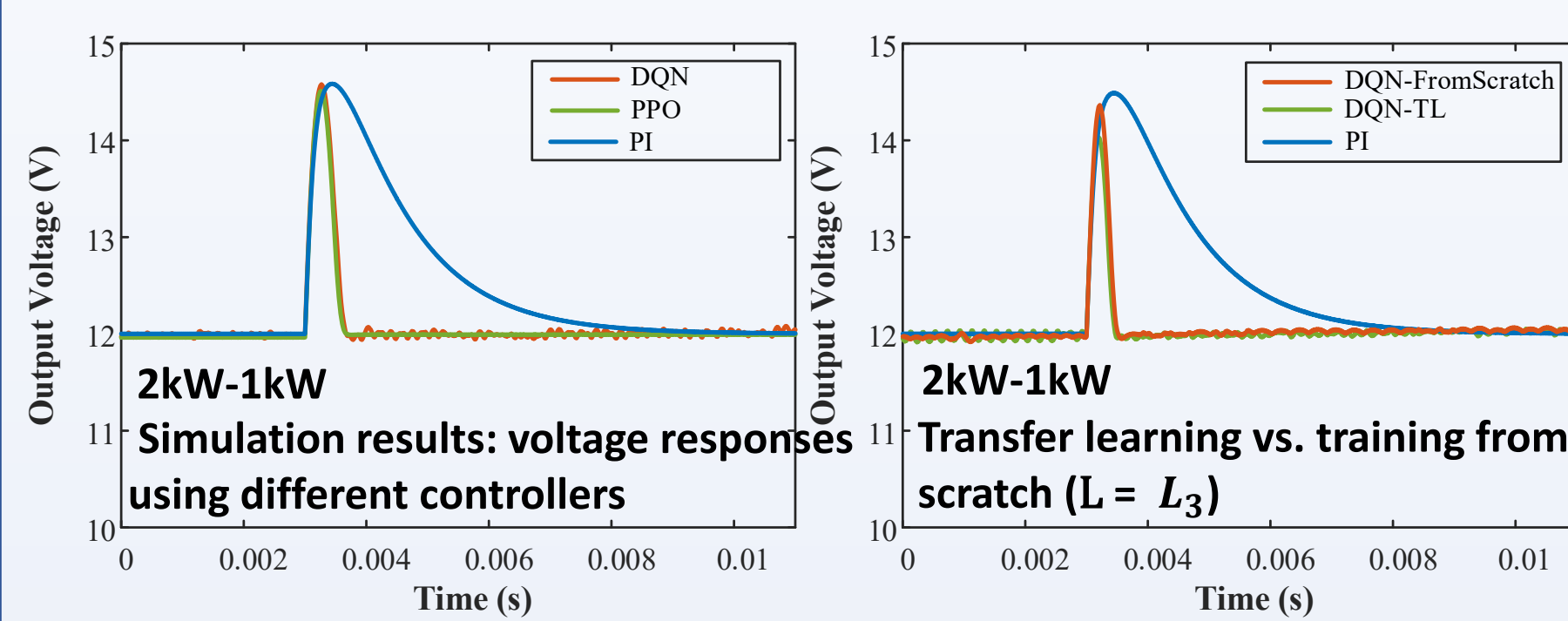
Require: Initialize the actor network $\pi_\theta(a_t|s_t)$ with parameters θ_0 and the critic network $V_\phi(s_t)$ with parameters ϕ_0 . Set the hyperparameters: discount factor γ , GAE factor λ , clipping factor δ , experience horizon H , mini-batch size b , and the number of epochs per update N .

- 1: **for** $k = 0, 1, 2, \dots$ **do**
- 2: Run the current policy π_{θ_k} in the MATLAB/Simulink environment.
- 3: At each sampling instant, observe the state s_t and sample the continuous duty cycle action $a_t = d_t$ from the Gaussian policy $\pi_{\theta_k}(a_t | s_t)$.
- 4: Apply d_t to the PWM module and collect the reward r_t and the next state s_{t+1} .
- 5: Store the collected trajectories as $\mathcal{D}_k$, where $\mathcal{D}_k = \{(s_t, a_t, r_t, s_{t+1})_{t=0}^{H-1}\}$.
- 6: Estimate the return $\hat{R}_t$ and the advantage $\hat{A}_t$ using the critic network $V_{\phi_k}(s_t)$ and GAE.
- 7: **for** $n = 1, 2, \dots, N$ **do**
- 8: Sample mini-batches from $\mathcal{D}_k$.
- 9: Update the actor parameters θ_k by maximizing the PPO clipped objective.
- 10: Update the critic parameters ϕ_k by minimizing the value-function loss.
- 11: **end for**
- 12: **end for**



DQN and PPO agents are trained for 200 episodes under different load-step conditions. PPO shows faster reward convergence during training. Compared with training from scratch, transfer learning(TL) reduces retraining effort and improves transient performance.

3. Simulation Verification

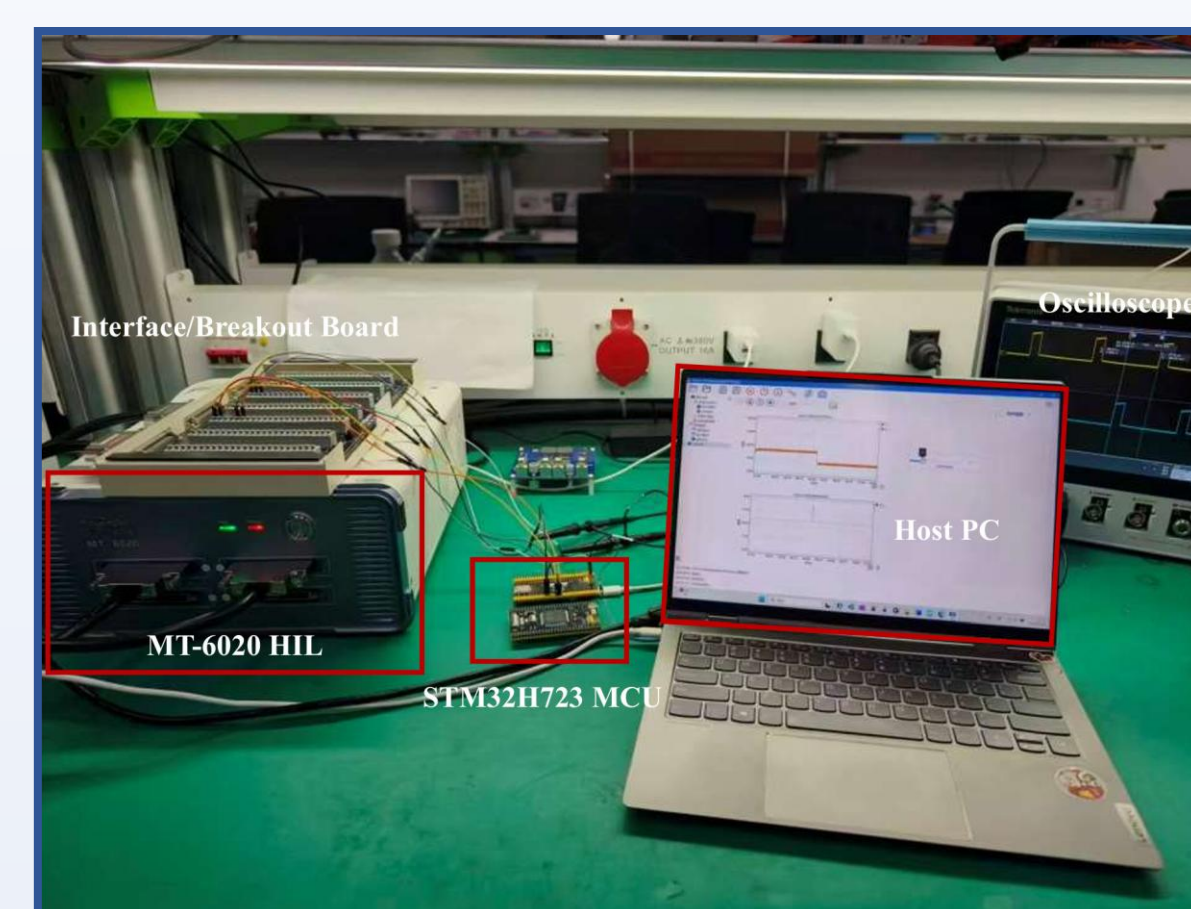


Simulation results show that both DRL controllers achieve faster voltage recovery than the PI controller.

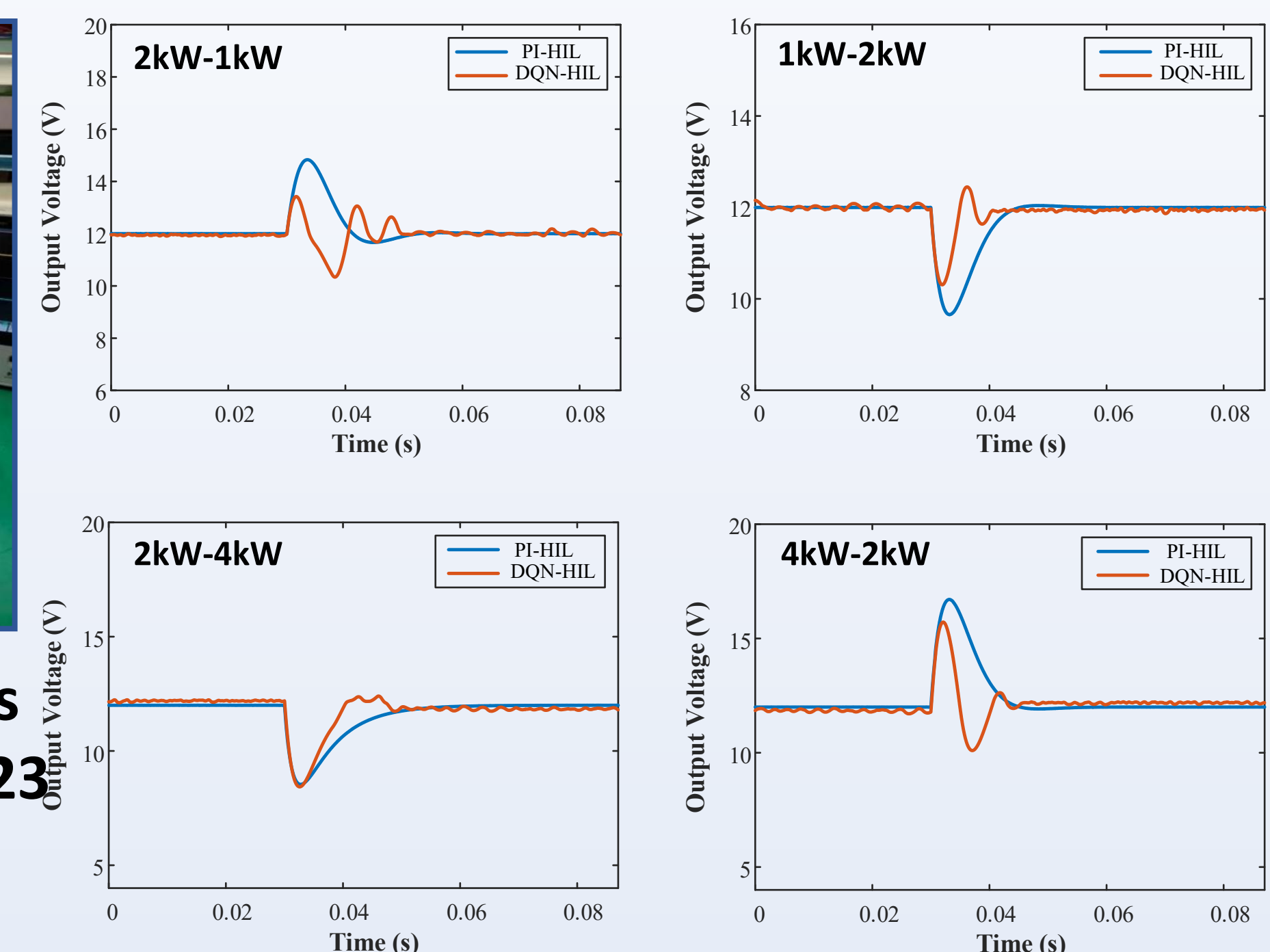
Controller	Case	Overshoot (V)	Settling time (ms)
PI	L_1	2.585	6.475
DQN	L_1	2.579	0.673
PPO	L_1	2.511	0.675
PI	L_3	2.491	6.364
DQN (From Scratch)	L_3	2.364	0.513
DQN (Transfer Learning)	L_3	2.033	0.469

PPO achieves slightly lower voltage overshoot than DQN under the nominal inductance case. With transfer learning, the DRL controller achieves lower overshoot and shorter settling time under inductance variations.

4. MCU Deployment and HIL Experiment



A lightweight DQN controller is implemented on an STM32H723 MCU for real-time inference.



The trained Q-network is converted from MATLAB/Simulink to a Keras model and imported into STM32CubeMX using X-CUBE-AI. The measured inference latency is 35 μ s, which is shorter than the 50 μ s PWM period. These results demonstrate the embedded feasibility of DRL-based power converter control.

5. Conclusion

This work develops a model-free DRL-based control framework for a two-phase interleaved buck converter using a train-test workflow. Future work will explore network compression to accelerate inference and extend the approach to higher-frequency converters.